

Tarek Mohamed

📍 Alexandria, Egypt ✉ tarekdx3@gmail.com ☎ 01274829005 in tarek-mohamed-salah 🔗 tarekdx34

Objective

Software Engineer (Frontend Developer) and senior Electronics and Communication Engineering student with proven experience building production-ready web applications. Skilled in React, TypeScript, JavaScript, HTML, and CSS with strong foundation in software architecture, API integration, and full-stack development principles. Successfully delivered commercial projects under tight deadlines while maintaining code quality and user experience standards. Seeking to leverage technical expertise and collaborative problem-solving abilities to contribute to innovative frontend development projects in a dynamic engineering environment.

Education

Alexandria University <i>BS in Communication and Electronics Engineering</i>	Alexandria, Egypt Sept 2021 – Sept 2026
◦ GPA: 3.41/4.0, Excellent ◦ Summary: Gained in-depth knowledge and practical experience in analog and digital circuit design, signal processing, transmission systems, and embedded systems. Developed strong technical skills through coursework and projects involving Embedded C, Assembly Language, and operating systems. ◦ Core Coursework: Embedded Systems, Operating Systems, Data Structures & Algorithms, Assembly Language, Embedded C, Python, C, Project Management	

Scholarships & Training

React Frontend Web Developer Scholarship <i>Digital Egypt Pioneers Initiative (DEPI) - Ministry of Communications and Information Technology</i>	Alexandria, Egypt (Hybrid) Jun 2025 – Dec 2025
◦ Selected for Egypt's Digital Pioneers Initiative (DEPI), Cohort 3 - an intensive 6-month scholarship program designed to develop job-ready skills in modern web development ◦ Building responsive, interactive web applications using React, TypeScript, and modern UI/UX design principles ◦ Learning full-stack development with Node.js, Express, and containerization using Docker and Kubernetes ◦ Working on comprehensive capstone project integrating all learned technologies ◦ Technologies: React, TypeScript, JavaScript, HTML5, CSS3, Node.js, Express, Git, GitHub, Docker, Kubernetes	

Experience

Software Engineer Intern <i>Alex Eagles Aero Design</i>	Alexandria, Egypt Sept 2024 – Present
◦ Developed Python scripts to enable autonomous drone movement and obstacle avoidance systems ◦ Implemented mapping techniques for terrain analysis and real-time decision-making in flight operations ◦ Contributed to problem-solving strategies for enhancing drone autonomy and operational efficiency ◦ Applied knowledge of embedded systems, aerodynamics, and software architecture in UAV control systems	
Software Engineer Intern <i>ALX</i>	Remote Feb 2023 – Feb 2024
◦ Completed intensive, hands-on training in software engineering fundamentals through real-world projects and peer collaboration ◦ Developed strong skills in system programming, data structures, and backend development with a DevOps mindset ◦ Key Skills: C Programming, Python, Data Structures & Algorithms, Linux/Unix Programming, Git & GitHub, Shell Scripting, Infrastructure Design, Backend Development, Problem Solving, Debugging	

Projects

Ajarly - Real Estate Rental Platform

2025

- Comprehensive real estate rental platform connecting property owners, brokers, and renters in Egypt
- Led frontend development and contributed significantly to backend architecture, building a production-ready system
- **Backend (Java + Spring Boot):** Implemented JWT authentication, property management, booking system with conflict handling, payment simulation, reviews, analytics with daily/monthly insights, and admin operations
- **Frontend (React + TypeScript):** Designed complete UI/UX, built reusable components with shadcn/ui and Radix, implemented state management and form validation, integrated search with dynamic filters
- **Key Features:** Multi-role system, property listings with Cloudinary, advanced search, booking/payment processing, owner analytics dashboard, admin panel, subscription plans
- **Technologies:** Java, Spring Boot, React, TypeScript, Tailwind CSS, JWT, Cloudinary

My Trip - Flight Booking and Management System

2025

- First paid frontend project - complete flight booking system from search to electronic ticket issuance
- Delivered production-ready system in 10-day timeline with dedicated portals for passengers, crew, front desk, and admin
- **Frontend Responsibilities:** Designed all interfaces, built reusable React components, integrated APIs, structured data flow and state management
- **Key Features:** Flight search with filters, booking/payment processing, electronic ticketing, cancellation handling, flight management, notifications, admin dashboard, role-based access
- **Technologies:** React.js, Tailwind CSS, Axios, React Router, Radix UI
- Collaborated with backend developers (Java Spring Boot)

Autonomous Drone System

Sept 2024 – Jun 2025

- Developing comprehensive autonomous drone system with focus on control, navigation, and mission planning
- Developed and tested drone control class using DroneKit-Python and MAVLink protocol
- Participated in backend-frontend integration for real-time mission updates and map visualization
- Collaborated in team of 6 developers on global path planning, SDK integration, and UI/UX design
- **Technologies:** Python, DroneKit-Python, MAVLink, JavaScript, React, Git

Technical Skills

Frontend: HTML5, CSS3, JavaScript (ES6+), React.js, TypeScript, Tailwind CSS, shadcn/ui, Radix UI, React Hook Form, Zod, Responsive Design (Flexbox, CSS Grid, Media Queries)

Backend & APIs: Java, Spring Boot, Node.js (learning), Express (learning), RESTful APIs, JWT Authentication, API Integration

Languages: JavaScript, TypeScript, Java, Python, C, Embedded C, Assembly Language

Tools: Git & GitHub, VS Code, Chrome DevTools, Docker (learning), Kubernetes (learning), MATLAB Simulink, LaTeX/Overleaf

Practices: Component-Based Architecture, State Management, UI/UX Principles, Responsive Design, Version Control, Agile

Languages & Soft Skills

Languages: English (Expert), Arabic (Native)

Soft Skills: Team Collaboration, Leadership & Communication, Problem Solving, Project Management, Time Management under Pressure